Stage 16N Inhomogeneous Plate-with-Hole Cycle-Jump Benchmark
Living Technical Report

Master Thesis Preparation

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Abstract

This living report records the geometry, numerical setup, HPC configuration, reference results, reinjection tests, diagnostic failures, and current conclusions for Stage 16N. The stage uses an inhomogeneous Abaqus plate-with-hole benchmark with a NEML-equivalent Chaboche UMAT. The full 1000-cycle non-jump reference has been completed and is used as the validation baseline for later fixed and adaptive cycle-jump simulations. The present state of the work is that the global response can be reproduced well after manual `SDVINI/SIGINI` state injection, but exact local state reproduction near the hole is not achieved by manual state initialization alone. Therefore later cycle-jump errors must be interpreted relative to the measured reinjection error floor.

Contents

1 Purpose and Current Status

The purpose of Stage 16N is to move from single-element or homogeneous cycle-jump demonstrations to an inhomogeneous finite-element benchmark. The model is deliberately more demanding: a plate with a central hole produces localized cyclic plasticity near the hole, so the method must be judged not only from global reaction-force loops but also from local stress and state-variable evolution.

Table 1: Current Stage 16N infrastructure and validation status.

Item	Status
1000-cycle full non-jump reference	Completed
16-CPU production policy	Locked
Abaqus threaded launcher	Verified
Adaptive ΔN table	Completed
Exact reinjection scaffold	Prepared
HPC mail policy	Fixed
B0-1 exact reinjection solver run	Completed
B0-1 scientific result	Practical baseline accepted
B0 initialization-only audit	Completed
B0-2 / B0-3 exact reinjection jobs	On hold
Fixed state-initialized jump validation	Scaffold prepared
Adaptive cycle-jump workflow	On hold

2 Geometry and Model Definition

The Stage 16N model is a three-dimensional plate-with-hole benchmark. It is intended to create localized cyclic evolution near the hole while retaining a global hysteresis response that can be compared across full-reference, exact-reinjection, fixed-jump, and adaptive-jump runs.

Table 2: Stage 16N model geometry and mesh summary.

Quantity	Value
Plate length	20.0
Plate height	10.0
Plate thickness	1.0
Central hole radius	1.0
Mesh spacing	$\Delta x = \Delta y = 0.25$
Element type	C3D8
Nodes	6642
Elements	3148
Hole-ring elements	60
Material model	NEML-equivalent Chaboche UMAT
State variables	27 SDVs

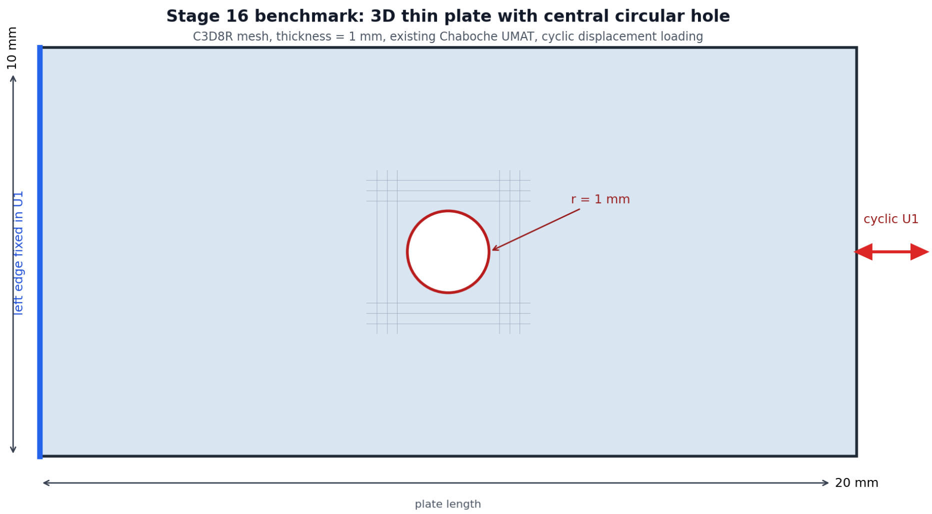

Figure 1: Plate-with-hole geometry used for the Stage 16N benchmark.

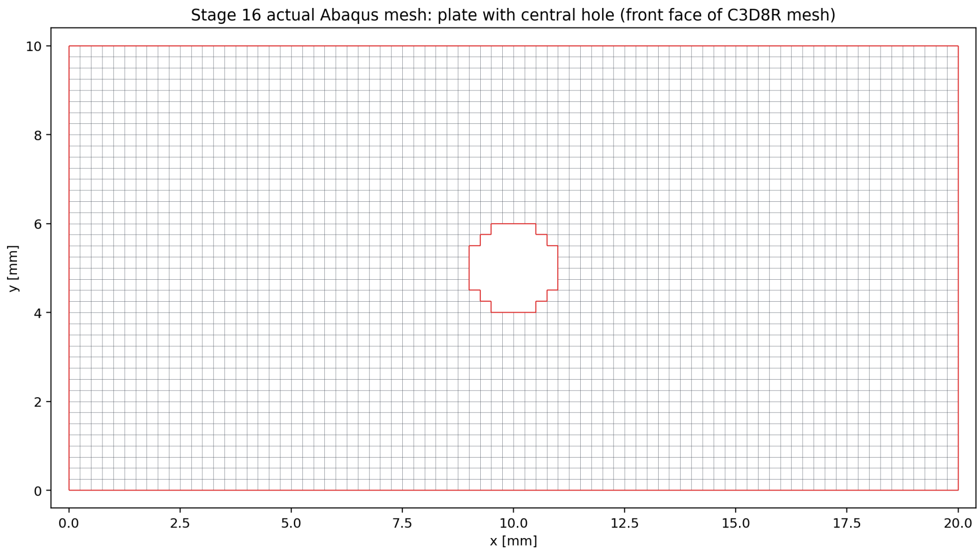

Figure 2: Actual structured C3D8 mesh with the element-wise approximation of the central hole.

2.1 Boundary Conditions and Loading

The left edge is fixed in the axial direction. Two anchor nodes constrain rigid-body modes. The right edge receives a cyclic axial displacement amplitude of ± 0.10. The full reference contains 1000 full non-jump cycles. Selected field output is stored at cycles 1, 2, 10, 50, 100, 250, 500, 750, and 1000.

3 HPC and Production Policy

The first corrected 1000-cycle parallel reference used 30 OpenMP threads, but PBS accounting showed that the average effective CPU usage was far below the requested 30 cores. A 16-CPU verification run was therefore used to lock a more resource-efficient production policy.

Table 3: Locked Stage 16N production setting.

Setting	Value
Production default	1 MPI rank × 16 OpenMP threads
PBS request	`select=1:ncpus=16:mpiprocs=1:ompthreads=16`
Abaqus launch	`cpus=16 mp_mode=threads`
Expected message-file line	`1 MPI RANK x 16 THREADS`

This policy is not claimed to perfectly saturate all 16 CPUs. It is a resource-efficient compromise that prevents the earlier serial fallback and avoids reserving 30 CPUs for a workload that did not use them efficiently.

3.1 Mail Notification Policy

Future PBS jobs must explicitly request mail on abort, begin, and end:

```
#PBS -m abe
#PBS -M pr21vyci@mailserver.tu-freiberg.de
```

This was added because PBS accounting previously showed `Mail_Points = a`, which only requests abort/failure mail.

4 Full 1000-Cycle Non-Jump Reference

The full non-jump reference was completed using Abaqus/Standard with the corrected threaded launcher.

Table 4: Completed 1000-cycle reference run.

Quantity	Value
PBS job	1335408.mmaster02
Abaqus job	`stage16n_parallel_max_reference_1000cycles`
Walltime	17:56:43
CPU time	178:03:42
Requested resources	30 CPUs, 135 GB
Abaqus parallelism	1 MPI rank × 30 threads
Completed cycles	1000
Last partial cycle	none

Table 5: Reference evolution from cycle 1 to cycle 1000.

Quantity	Cycle 1	Cycle 1000	Change
`RF1_max`	2356.834	2908.244	+23.40%
`RF1_min` magnitude	2509.712	3074.823	+22.52%
`loop_area_abs`	433.822	577.068	+33.02%
`HOLE_RING_MISES_MAX`	398.503	492.827	+23.67%
`HOLE_RING_SDV1_MAX`	0.0561	26.0519	–
`HOLE_RING_SDV8_MAX`	0.2996	92.4455	–
`HOLE_RING_SDV11_MAX`	7.9019	61.3076	–

The evolution clearly exceeds the original continuation thresholds for global loop quantities and local hole-ring state variables. This confirms that the plate-with-hole benchmark is meaningful for cycle-jump validation.

5 Adaptive ΔN Table

The adaptive ΔN table was computed from the completed 1000-cycle reference. It uses dense global cycle metrics and selected local field-output anchors. The controlling variable is the most sensitive monitored quantity, not only the global reaction force.

Table 6: Conservative adaptive ΔN estimates from the 1000-cycle reference.

Base cycle	Next anchor	Target	ΔN	Controlling variable
1	2	2	1	HOLE_RING_SDV8_MAX
2	10	3	1	HOLE_RING_SDV8_MAX
10	50	14	4	HOLE_RING_SDV8_MAX
50	100	61	11	HOLE_RING_SDV1_MAX
100	250	124	24	HOLE_RING_SDV1_MAX
250	500	299	49	HOLE_RING_SDV1_MAX
500	750	575	75	HOLE_RING_SDV1_MAX
750	1000	849	99	HOLE_RING_MISES_MAX

The table shows that local state variables near the hole strongly restrict the allowable jump size. This supports the decision that any later fixed/adaptive jump validation must include local metrics, not only global RF loops.

5.1 Reinjection-Aware Controller Update

After the B0 initialization audit, `HOLE_RING_SDV8_MAX` was downgraded from a hard pass/fail controller to a diagnostic-only variable because it already shows a large mismatch immediately after manual `SDVINI`/`SIGINI` initialization. A reinjection-aware table was therefore generated with `SDV8` retained in diagnostic output but removed from the hard controller list.

Table 7: Reinjection-aware adaptive ΔN estimates with `SDV8` diagnostic-only.

Base	Next anchor	Target	ΔN	Controller
1	2	2	1	HOLE_RING_SDV11_MAX
2	10	3	1	HOLE_RING_SDV1_MAX
10	50	15	5	HOLE_RING_SDV1_MAX
50	100	61	11	HOLE_RING_SDV1_MAX
100	250	124	24	HOLE_RING_SDV1_MAX
250	500	299	49	HOLE_RING_SDV1_MAX
500	750	575	75	HOLE_RING_SDV1_MAX
750	1000	849	99	HOLE_RING_MISES_MAX

The first conservative fixed-jump gate remains cycle 100 to cycle 125, followed by normal continuation to cycle 250. This row is still controlled by `HOLE_RING_SDV1_MAX`, not by diagnostic-only `SDV8`.

6 Stage 16N-B: State-Initialized Continuation Verification

Before any real cycle jump is allowed, the workflow must prove that stress and state-variable fields can be transferred into a new Abaqus run. The route used here is manual state-initialized continuation through `SDVINI/SIGINI`; it is not a native Abaqus restart. The originally intended verification cases were:

- B0-1: exact cycle 100 state, continue normally to cycle 250.
- B0-2: exact cycle 250 state, continue normally to cycle 500.
- B0-3: exact cycle 500 state, continue normally to cycle 1000.

B0-2 and B0-3 remain on hold. The B0-1 and initialization-audit results are now treated as the measured reinjection baseline rather than as proof of an exact local restart.

6.1 State Reader Implementation

The initial CSV preload design failed under threaded Abaqus because multiple threads attempted to load the same file concurrently. The state-transfer implementation was therefore changed to a direct-access binary reader. Each record contains:

```
S1, S2, S3, S4, S5, S6, SDV1, ..., SDV27
```

The record number is:

```
record = (NOEL - 1) * 8 + NPT
```

On the Abaqus/Intel Fortran environment used here, direct-access `RECL` is interpreted in 4-byte words. Therefore the correct value for 33 double-precision values is:

```
RECL = 66
```

7 B0-1 Cycle-100 to Cycle-250 State-Initialized Continuation

B0-1 completed the Abaqus solver successfully. PBS reported exit status 2 only because the original postprocessing path failed after solver completion; extraction was rerun manually afterward.

Table 8: B0-1 run accounting and solver status.

Quantity	Value
PBS job	1336497.mmaster02
Route	cycle 100 state $\rightarrow$ cycle 250
PBS walltime	01:41:36
PBS CPU time	10:52:42
Requested CPUs	16
Abaqus solver	completed
Abaqus parallelism	1 MPI rank $\times$ 16 threads
Scientific status	Practical baseline accepted

Table 9: B0-1 comparison against the 1000-cycle reference at cycle 250.

Quantity	Relative error
RF1_max	0.259048%
RF1_min	0.687538%
loop_area_abs	0.967300%
HOLE_RING_MISES_MAX	7.63648%
HOLE_RING_S11_MAX_ABS	9.58753%
HOLE_RING_SDV1_MAX	0.360172%
HOLE_RING_SDV8_MAX	10.9164%
HOLE_RING_SDV11_MAX	7.27213%

The global response is excellent, but the local hole-ring stress and state-variable errors are too large for a clean exact-restart claim.

8 B0-1 Local Diagnostic Review

A pointwise hole-ring diagnostic was run to determine whether the high local error was only a maximum-location artifact. It was not. The reference and reinjection maxima for `SDV8` occurred at the same element and integration point.

Table 10: B0-1 SDV8 argmax diagnostic at cycle 250.

Quantity	Value
Reference SDV8 argmax	element 1242, IP 7
Reinjection SDV8 argmax	element 1242, IP 7
Same argmax location	true
Reference value	91.1466598511
Reinjection value	81.1967315674
Argmax relative error	10.9164%

This made B0-1 a real local mismatch rather than a harmless maximum-location shift.

9 B0 Cycle-100 Initialization-Only Audit

The initialization audit was created to answer whether the local mismatch already exists immediately after exact cycle-100 initialization, before any cycle-100 to cycle-250 continuation.

Table 11: Cycle-100 binary reader and initialization audit.

Quantity	Value
CSV-vs-binary maximum absolute error	3.55×10^{-15}
Common hole-ring element/IP records	480
`SDV8` median relative error	12.7022%
`SDV8` 95th percentile relative error	150.665%
`SDV8` maximum relative error	479.090%
`SDV8` argmax location	same element/IP, 1242/7

The binary extraction and reader pipeline is correct to machine precision. Therefore the local mismatch is not caused by the Python binary writer or direct-access file format. The mismatch appears during manual Abaqus initialization and the tiny equilibrium step.

9.1 Initialization Audit Interpretation

The most likely interpretation is that `SDVINI/SIGINI` inject stress and state variables correctly, but the manual state-initialized model does not reconstruct the compatible displacement, strain, solver-history, and full equilibrium state of the original reference analysis. Strain-like local state variables such as `SDV8` are therefore adjusted during the first equilibrium solve. This means B0-1 is useful as a practical state-initialized continuation test, but it is not a mathematically exact Abaqus restart for local fields.

10 Reinjection Error Floor and Revised Metric Policy

The B0 audit closes Stage 16N-B with the following conclusion:

> Global continuation works, binary state transfer works, but local exact restart is not achieved with `SDVINI/SIGINI` alone.

Future fixed and adaptive jump results must separate the manual reinjection error floor from the additional error caused by cycle-jump extrapolation. The primary pass/fail metrics are `RF1` maximum/minimum, loop area, global hysteresis shape, `HOLE_RING_SDV1_MAX`, `HOLE_RING_MISES_MAX`, and `HOLE_RING_S11_MAX_ABS`. `HOLE_RING_SDV8_MAX` remains important but is diagnostic-only because it is already affected by initialization before any cycle-jump approximation is applied.

11 Stage 16N-C Fixed State-Initialized Jump Scaffold

Stage 16N-C now begins with one conservative fixed jump:

Table 12: First fixed state-initialized jump case.

Quantity	Value
Case name	`B1_100_to_125_to_250`
Base state	cycle 100
Interpreted jump target	cycle 125
Normal Abaqus continuation	cycles 126–250
Comparison endpoint	cycle 250
Initial state strategy	zero-order hold from cycle 100
Baseline for interpretation	B0-1 cycle 100 to 250 continuation

The zero-order state strategy is intentionally conservative. It measures the error added by skipping cycles 101–125 on top of the known manual reinjection floor. If this case gives acceptable global error and controlled primary local errors, the next fixed jump cases can be expanded to cycle 250 to 300, cycle 500 to 575, and cycle 750 to 850.

12 Failures and Fixes

Table 13: Important failures and fixes recorded in Stage 16N.

Issue	Observation	Fix / conclusion
Accidental serial-like run	A 22-hour run reached only 592 full cycles.	Parallel launcher corrected; later 1000-cycle reference completed.
30-CPU over-allocation	30-thread run averaged about 10 effective cores.	16 CPUs selected as production compromise.
Missing start/end email	PBS accounting showed `Mail_Points = a`.	Added `#PBS -m abe` and mail user to Stage 16N submit scripts.
CSV state-reader race	Threaded Abaqus calls attempted concurrent CSV preload.	Replaced with direct-access binary state reader.
Wrong direct-access record length	Abaqus/Intel runtime expected word-based `RECL`.	Set `RECL = 66` for 33 double values.
Postprocessing path failure	B0-1 solver completed but PBS exit status was 2.	Fixed extractor path and reran extraction manually.
B0-1 local mismatch	Global errors under 1%, local `SDV8` max error 10.9164%.	Added pointwise diagnostic and initialization-only audit.
Initialization local mismatch	Binary audit passed, but `SDV8` mismatch appeared immediately.	Concluded manual `SDVINI/SIGINI` is not a mathematically exact local restart.
Metric policy mismatch	Original adaptive table allowed `SDV8` to act as a hard controller.	Added reinjection-aware table with `SDV8` diagnostic-only.

13 Current Conclusion

Stage 16N has successfully produced a full 1000-cycle non-jump reference and a conservative adaptive ΔN table. The inhomogeneous plate-with-hole benchmark is scientifically useful because both global hysteresis quantities and local hole-ring state variables evolve significantly. The HPC workflow is now resource-controlled with a 16-CPU production policy and explicit start/end/failure mail notifications.

The important methodological limitation is exact local reinjection. B0-1 proves that the manual `SDVINI/SIGINI` workflow can run and reproduce the global force-displacement response well, but it does not reproduce local strain-like state variables exactly. The initialization-only audit shows that this local mismatch appears immediately during manual Abaqus initialization/equilibration, even though the binary reader is exact.

14 Next Work

The next work is to run the first conservative Stage 16N-C fixed state-initialized jump and interpret it against the B0-1 reinjection baseline. B0-2 and B0-3 remain on hold as exact-local-restart claims, but practical fixed-jump validation can proceed with the reinjection floor explicitly documented.

1. Prepare and submit `B1_100_to_125_to_250` on HPC using the locked 16-CPU production policy.

2. Compare the fixed-jump result against the 1000-cycle reference and against the B0-1 reinjection baseline.

3. Keep `SDV8` as a diagnostic warning metric, not a hard pass/fail metric.

4. If the first case is acceptable, expand to larger fixed state-initialized jumps.

A Key Repository Evidence Files

- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_parallel_ma`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_adaptive_de`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_adaptive_de`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_REINJECTION`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_FIXED_JUMP_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_HPC_MAIL_PO`